

TechHMS – Enterprise Hospital Management System

Module 8 – Pharmacy Management

Version: 1.0

1. Module Overview

The **Pharmacy Management Module** manages the complete medication lifecycle—from procurement and inventory management to prescription validation, dispensing, billing, stock reconciliation, controlled drug tracking, and expiry management.

The module integrates with EMR, OPD, IPD, Purchase, Inventory, Billing, Finance, and Insurance to ensure accurate medication management, patient safety, and financial accountability.

2. Objectives

The Pharmacy module aims to:

- Digitize pharmacy operations.
- Ensure safe medication dispensing.
- Prevent medication errors.
- Maintain accurate inventory.
- Support batch and expiry tracking.
- Enable barcode-based dispensing.
- Optimize medicine procurement.
- Integrate seamlessly with EMR and Billing.
- Support regulatory compliance for controlled substances.

3. User Roles

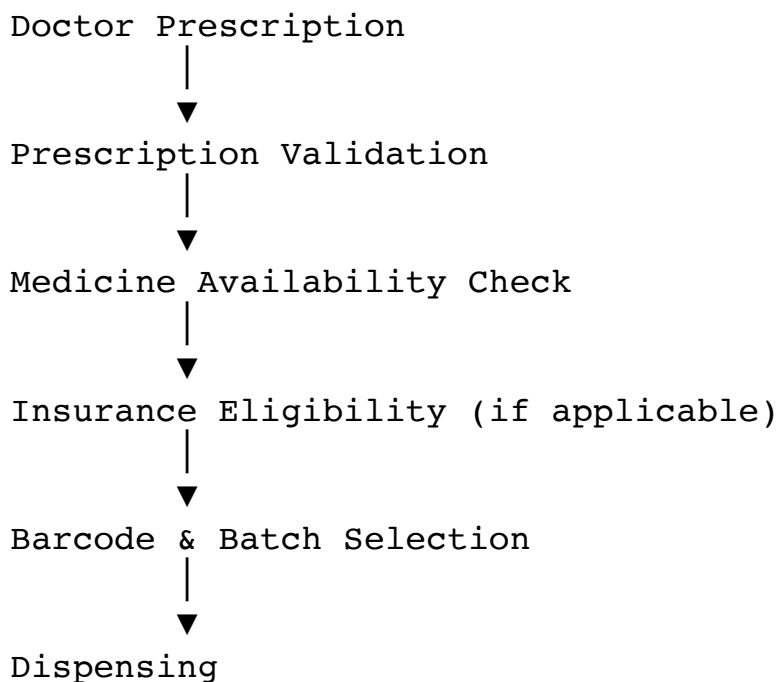
- Pharmacist
- Senior Pharmacist
- Pharmacy Manager
- Store Keeper
- Purchase Manager
- Billing Executive
- Doctor (Prescription Only)
- Nurse (Medication Administration View)
- Finance Manager
- Hospital Administrator

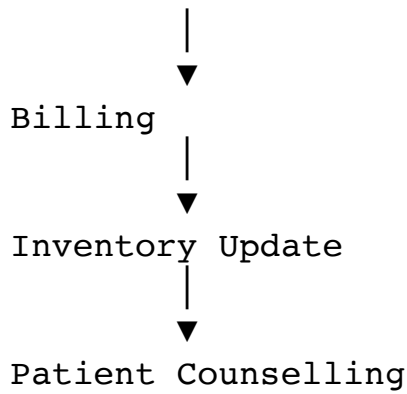
4. Functional Scope

The Pharmacy module includes:

- Medicine Master
- Generic & Brand Management
- Drug Classification
- Batch & Expiry Management
- Barcode Management
- Purchase Integration
- Goods Receipt Integration
- Stock Transfers
- Prescription Validation
- Outpatient Dispensing
- Inpatient Dispensing
- Emergency Pharmacy
- Controlled Drug Register
- Returns & Refunds
- Expired Medicine Handling
- Stock Adjustment
- Physical Stock Verification
- Reorder Management
- Billing Integration
- Insurance Medication Processing
- Vendor Management
- Pharmacy Reports
- Audit Trail

5. Pharmacy Workflow





6. Pharmacy Categories

TechHMS supports multiple pharmacy types:

- Outpatient Pharmacy (OP)
- Inpatient Pharmacy (IP)
- Emergency Pharmacy
- Operation Theatre Pharmacy
- ICU Pharmacy
- Retail Pharmacy
- Satellite Pharmacy
- Central Medical Store

7. Medicine Master

Each medicine contains:

Basic Information

- Medicine Code
- Generic Name
- Brand Name
- Manufacturer
- Strength
- Dosage Form
- Route
- ATC Classification
- Drug Category

Pricing

- Purchase Price
- Selling Price
- MRP
- Insurance Price
- GST / Tax

Inventory

- Reorder Level
- Maximum Stock
- Minimum Stock
- Storage Temperature
- Shelf Life

Regulatory

- Prescription Required
- Narcotic Drug
- Controlled Drug
- High Alert Medication

8. Batch Management

Every receipt records:

- Batch Number
- Manufacturing Date
- Expiry Date
- Supplier
- Purchase Price
- Quantity
- Barcode
- Storage Location

The system always dispenses based on **FEFO (First Expiry First Out)** unless overridden by an authorized pharmacist.

9. Barcode Management

Supported barcode standards:

- EAN-13
- Code 128
- QR Code
- GS1 (Future)

Barcode scanning supports:

- Goods Receipt
- Dispensing
- Stock Transfer
- Stock Verification
- Returns

10. Prescription Validation

Before dispensing, the system verifies:

- Prescription authenticity
- Doctor authorization
- Drug availability
- Quantity
- Allergy conflicts
- Drug interactions
- Duplicate therapy
- Expired prescriptions
- Controlled drug restrictions

11. Outpatient Dispensing

Workflow:

1. Search patient.
2. Retrieve electronic prescription.
3. Validate medicine availability.
4. Reserve stock.
5. Scan barcode.
6. Dispense medicine.
7. Generate bill.
8. Print labels and instructions.
9. Update inventory.

12. Inpatient Dispensing

Supports:

- Unit Dose
- Ward Stock
- Floor Stock
- Daily Medication Supply
- STAT Medication
- Emergency Medication

The system integrates with the Medication Administration Record (MAR) to track medication issuance and administration.

13. Controlled Drug Management

Features include:

- Separate controlled drug register.
- Dual authorization for issue and return.
- Daily balance verification.
- Mandatory reason for adjustments.
- Regulatory reporting.

14. Returns & Refunds

Supports:

- OP Returns
- IP Returns
- Damaged Medicines
- Expired Medicines
- Supplier Returns

All returns require authorization and update inventory accordingly.

15. Stock Management

Functions:

- Stock Transfer Between Stores
- Stock Adjustment
- Cycle Counting
- Physical Stock Verification
- Stock Reservation
- Stock Blocking
- Inter-branch Transfers

16. Expiry & Near-Expiry Management

The system generates alerts for medicines nearing expiry (configurable periods such as 30, 60, or 90 days).

Users can:

- Block dispensing of expired medicines.
- Initiate supplier return requests.
- Transfer stock to high-consumption locations.
- Generate destruction records.

17. Purchase Integration

The Pharmacy module integrates with the Purchase module to:

- Generate purchase requisitions based on reorder levels.
- Receive goods through GRN.
- Validate invoices.
- Update inventory.
- Record supplier batch details.

18. Billing Integration

Automatic billing for:

- OP medicines
- IP medicines
- Emergency issues
- Insurance-covered medicines
- Corporate billing

Supports:

- Discounts
- Promotional pricing
- Insurance co-pay
- Tax calculations

19. Insurance Processing

The system validates:

- Covered medicines.
- Quantity limits.
- Formulary restrictions.
- Prior authorization requirements.
- Patient co-payment.

20. Business Rules

- Expired medicines cannot be dispensed.
- FEFO is the default issue method.
- Controlled drugs require additional authorization.
- Stock cannot become negative unless explicitly permitted.
- Barcode validation is mandatory for configured medicines.
- Prescription modifications require clinician approval after dispensing begins.

21. Reports

- Daily Dispensing Register
- Medicine Consumption Report
- Controlled Drug Register
- Near Expiry Report
- Expired Stock Report
- Reorder Report
- Slow-Moving Medicines
- Fast-Moving Medicines
- Stock Valuation Report
- Vendor-wise Purchase Report

- Pharmacy Revenue Report
- Insurance Medicine Utilization
- Drug Recall Report

22. Dashboard Widgets

Pharmacy Dashboard

- Pending Prescriptions
- Low Stock Alerts
- Near Expiry Medicines
- Out-of-Stock Items
- Controlled Drug Balance
- Daily Dispensing Count
- Pharmacy Revenue
- Purchase Requisitions Pending

Management Dashboard

- Inventory Value
- Medicine Consumption Trends
- Top Dispensed Medicines
- Stock Turnover Ratio
- Expiry Loss Analysis
- Supplier Performance
- Gross Margin by Drug Category

23. High-Level Database Tables

- MedicineMaster
- MedicineCategory
- MedicineBatch
- PharmacyStock
- PharmacyStore
- PharmacyDispense
- Prescription
- PrescriptionItem
- ControlledDrugRegister
- StockTransfer
- StockAdjustment
- GoodsReceipt
- PharmacyReturn
- ReorderRequest
- PharmacyAudit

24. Security & Audit

- Role-based access for dispensing, stock adjustments, and pricing.

- Electronic approval for controlled drug transactions.
- Audit trail for all inventory movements and prescription activities.
- Support for configurable approval workflows for high-value adjustments.

25. Future AI Enhancements

TechHMS Pharmacy will leverage AI to improve medication safety and inventory efficiency:

- AI-driven demand forecasting based on seasonal trends, disease patterns, and historical consumption.
- Predictive reorder recommendations to reduce stockouts and excess inventory.
- AI alerts for unusual prescribing patterns and potential medication misuse.
- Intelligent substitution suggestions when prescribed medicines are unavailable, based on approved formularies.
- Detection of potential dispensing errors before issue.
- Automated identification of slow-moving and obsolete inventory.
- Personalized medication adherence reminders through the patient portal and mobile app.
- Executive analytics for pharmacy profitability, wastage, and procurement optimization.